The University of Texas at Austin
Solutions to In-Term Exam 2

M339D Introduction to Financial Mathematics

November 09, 2026

Instructions: This is a closed book and closed notes exam. The maximal score on this portion of the exam is 65 points.
Time: 50 minutes

Definition

Problem 2.1. (5 points) Write the definition of a **replicating** portfolio of a European option.

Problem 2.2. (5 points) Consider the following constants, in our usual notation,

$$u = 1.04, \quad d = 1.01, \quad h = 1/3, \quad r = 0.06.$$

Then, the no-arbitrage condition for the binomial asset pricing model is satisfied. *True or false? Why?*

Problem 2.3. (5 points) The realized returns on the continuously compounded bases are *additive*. *True or false? Why?*

FREE-RESPONSE PROBLEMS

Problem 2.4. (20 points) Consider a non-dividend-paying stock whose current price is \$90 per share. The stock has the volatility equal to 0.20.

Let the be equal to 0.03.

You model the evolution of this stock over the next year with a three-period **forward** binomial tree.

What is the price of a \$95-strike, one-year put on the above stock consistent with this model?

Problem 2.5. (10 points) A zombie apocalypse is in full swing. A human survivor has a star launcher with 100 silver stars and is shooting at zombies. The probability of successfully eliminating a zombie in a single throw is 0.80. Using the *normal approximation to the binomial*, what is the probability that the survivor will successfully eliminate at least 84 zombies?

Problem 2.6. (10 points) Let the continuously compounded risk-free interest rate equal 2%. The current price of an index equals \$1,000. The current premium on a six-month, \$970-strike 6-month call on this index is \$109.20. What is the price of an otherwise identical put option?

Problem 2.7. (5 points) Assume that $Y = e^X$ where X is a normal random variable with mean 3 and standard deviation 1. Find the mean of Y.

Hint: It helps if you use the expression for the moment generating function of a standard normal random variable.